CASE WHEN

Polymorphic Conditional Switchboard & Branch Router

SQL FOUNDRY // ARCHITECTURE MANUAL

In-Flight Scalar Evaluation (Usable in SELECT, WHERE, GROUP BY, ORDER BY)

Complexity: O(1) per row across K conditional branches.

1. ANSI Syntax & Grammar Specification

```
CASE [ expression ]
    WHEN condition THEN result
    [ WHEN ... THEN ... ]
    [ ELSE result ]
END
```

The CASE expression provides declarative conditional branching. It evaluates predicates sequentially and returns the corresponding result expression for the first condition that evaluates to TRUE. If no condition matches and an ELSE clause is omitted, the expression returns NULL.

2. Database Engine Internals & Storage Footprint

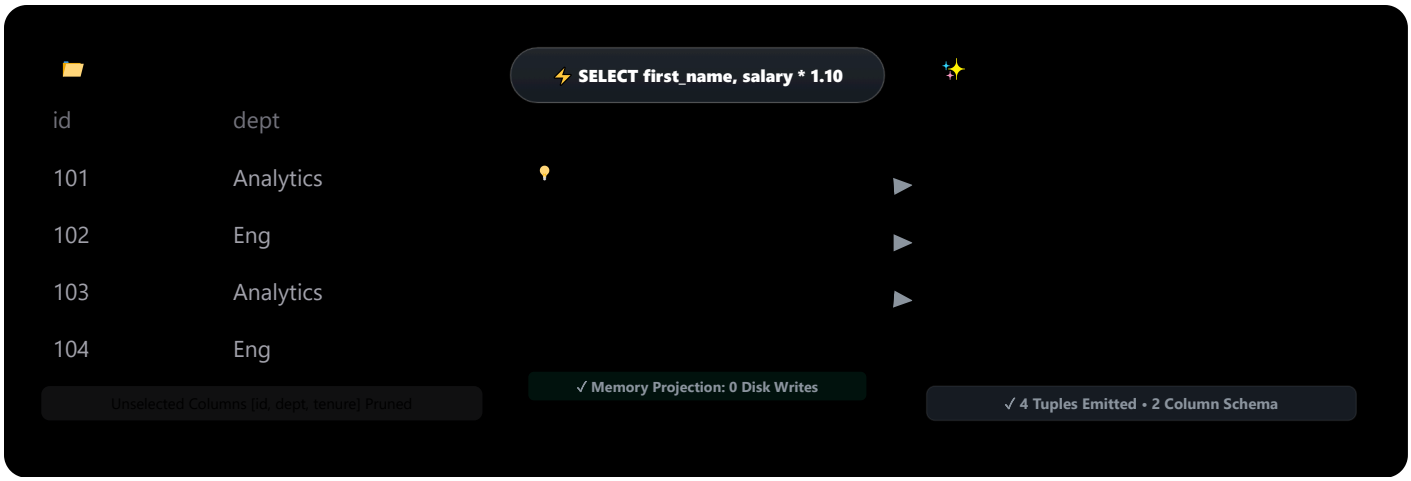
Physical Operator: Short-Circuit Scalar Evaluator

Memory Footprint: O(1) memory register.

Disk Spill Risk: None. Evaluated in CPU branch registers.

Index SARGability: Enables Conditional Aggregation (Pivot emulation via SUM(CASE WHEN ...)) without multiple table scans.

3. Relational Architecture Diagram (Tuple Lifecycle)



Overview | 1. Scan Table | 2. Relational Engine | 3. Emit Result

CASE WHEN // IMPLEMENTATION PATTERNS

MODULE: CONDITIONAL & PIPELINES

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4. Canonical Query Code Progression

Pattern A: Simple Searched CASE Expression

```
-- Categorizing employee compensation into corporate bands
SELECT
    emp_id,
    first_name,
    salary,
    CASE
        WHEN salary >= 120000 THEN 'SENIOR_BAND'
        WHEN salary >= 90000 THEN 'MID_BAND'
```

```
        ELSE 'ASSOCIATE_BAND'
    END AS compensation_tier
FROM enterprise_employees;
```

Pattern B: Conditional Aggregation (Fast Pivot Pattern)

```
-- Pivoting rows to columns in a single table scan
SELECT
    department,
    COUNT(*) AS total_headcount,
    SUM(CASE WHEN salary >= 100000 THEN 1 ELSE 0 END) AS high_earners_count,
    SUM(CASE WHEN salary < 100000 THEN 1 ELSE 0 END) AS standard_earners_count
FROM enterprise_employees
GROUP BY department;
```

Pattern C: Conditional Sorting (Dynamic Custom Ordering)

```
-- Custom domain sorting prioritizing VIP orders and delivery deadlines
SELECT order_id, customer_tier, promised_delivery_date, total_amount
FROM customer_orders
ORDER BY
    CASE customer_tier
        WHEN 'PLATINUM' THEN 1
        WHEN 'GOLD' THEN 2
        WHEN 'SILVER' THEN 3
        ELSE 4
    END ASC,
    promised_delivery_date ASC;
```

5. Production Schemas & Input/Output Relational Tables

Input: Input Salary Stream

name	salary
David	$125,000
Ashley	$115,000
Julia	$95,000
Carlos	$72,000

Emitted Output Table

name	salary	tier
David	$125,000	Senior Band
Ashley	$115,000	Mid Band
Julia	$95,000	Mid Band
Carlos	$72,000	Junior Band

6. Production Enterprise Case Study

Company / Team: Amazon Prime Entertainment

Operational Scenario: Risk models score user engagement based on weekly viewing minutes, watchlist saves, and app launch frequencies, assigning retention risk flags.

```
SELECT
    user_id,
    weekly_stream_minutes,
    CASE
        WHEN weekly_stream_minutes >= 300 AND watchlist_saves >= 5 THEN 'POWER_ENGAGED'
```

```
        WHEN weekly_stream_minutes >= 60 THEN 'MODERATE_ENGAGED'
        WHEN weekly_stream_minutes > 0 THEN 'AT_RISK_LOW_USAGE'
        ELSE 'DORMANT_CRITICAL'
    END AS retention_health_status
FROM prime_video_weekly_engagement;
```

Business & Infrastructure Impact: Classifies 85 million active streaming profiles nightly in a single scan for personalized homepage re-engagement algorithms.

7. FAANG & Wall Street Technical Interview Defense

[Meta / Netflix] Question 1:

Does CASE WHEN in SQL short-circuit like an if-else in Python/C++?

Model Answer: In general ANSI semantics, CASE expressions evaluate conditions sequentially and stop at the first matching condition. However, SQL query optimizers reserve the right to reorder expression evaluation for performance if no side effects exist. Never rely on CASE short-circuiting to prevent division by zero in deeply nested optimizer trees; use NULLIF() instead.

[Bloomberg] Question 2:

What is Conditional Aggregation, and why is it preferred over running multiple separate queries with WHERE filters?

Model Answer: Conditional aggregation wraps CASE WHEN inside SUM() or COUNT() (e.g. SUM(CASE WHEN status = "ERROR" THEN 1 ELSE 0 END)). It allows calculating metrics for multiple independent subsets in a single pass over the table, whereas separate queries require scanning the table multiple times.

CASE WHEN // DEFENSE AUDIT & MCQS

3VL TRAPS • CERTIFICATION AUDIT

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8. High-Yield Pitfalls & Corporate Production Bugs

High **Omitting the ELSE Clause:** If no WHEN branch matches and ELSE is omitted, CASE returns NULL. In mathematical calculations, this can introduce unexpected NULL values that break arithmetic.

Critical **Incompatible Return Data Types:** All THEN and ELSE result branches must resolve to compatible data types. Returning an integer in one branch and a string in another triggers immediate compile-time type coercion failures.

9. Certification & Conceptual Mastery Examination (5 MCQs)

1. What does a CASE expression return if no WHEN condition matches and no ELSE branch is specified?

A) 0
B) An empty string ''
C) NULL
D) A fatal runtime exception

Correct Answer: C • *ANSI SQL specifies that an omitted ELSE clause defaults to ELSE NULL.*

2. What is the result of: CASE WHEN 1=1 THEN 'A' WHEN 2=2 THEN 'B' ELSE 'C' END?

A) 'A'
B) 'B'
C) 'A, B'
D) 'C'

Correct Answer: A • *CASE evaluates sequentially and short-circuits on the first condition that evaluates to TRUE (1=1).*

3. How do you rewrite "COUNT(CASE WHEN status = 'PAID' THEN 1 END)" using SUM?

A) SUM(status = 'PAID')
B) SUM(CASE WHEN status = 'PAID' THEN 1 ELSE 0 END)
C) SUM(PAID)
D) SUM(1)

Correct Answer: B • *In SUM, non-matching rows must evaluate to 0 so they do not add to the accumulator.*

4. Can you use a CASE expression inside a GROUP BY clause?

A) No, only column names can be in GROUP BY

B) Yes, to dynamically bucket rows into custom groups (e.g. GROUP BY CASE WHEN age < 30 THEN 'Young' ELSE 'Senior' END)

C) Only in MySQL

D) Only if wrapped in a VIEW

Correct Answer: B • *Any deterministic scalar expression can be used as a grouping key in ANSI GROUP BY.*

5. What error occurs in: "SELECT CASE WHEN flag = 1 THEN 100 ELSE 'N/A' END FROM t"?

A) Syntax error at flag

B) Type mismatch error: THEN returns an integer while ELSE returns a varchar

C) Table scan error

D) No error; SQL dynamically types the column

Correct Answer: B • *All branches of a CASE expression must return coercible, identical data types.*